Project Overview

RoadBuddy is going to utilize a driver’s smartphone to act as a dashcam to detect road hazards real-time. The software will alert and possibly reroute other nearby drivers to keep them safe from road hazards.

Scope

- **In Scope**
  - SYS-01: Pothole Detection
    - To be useful, our system must detect road hazards, and potholes are a very common road hazard that computer vision models are already trained to detect.
  - SYS-02: Map Rerouting
    - Our system must interact with a mapping API to allow users to safely avoid a detected road hazard if severe enough.
  - SYS-03: Hazard Classification
    - In order to understand whether a user should be rerouted or just simply audibly warned of an upcoming hazard, our system must classify hazards based on researched potential damage.
  - SYS-04: Real-Time
    - Our system must be able to respond in real-time, so that other drivers are alerted as soon as possible of upcoming hazards.
  - SYS-05: Accurate Reports
    - Our system must have a high accuracy rate on its classification of hazards so that it does not inconvenience drivers and push them away from the app.
  - SYS-06: Clear Weather Conditions
    - Our system must have all these requirements for clear weather, sunny days.
- **Under Evaluation**
  - SYS-07: All Weather Conditions
    - It may take a considerable amount of time to find training data for more weather conditions, but it would make the app a lot more useful.
  - SYS-08: More Hazards Detection
    - This can include anything that has entered the roadway that is not supposed to be there, including tires, couches, mattresses, fallen trees, etc. There are pre-trained models that can do this, so we are still considering it.
  - SYS-09: Emergency Vehicles Detection
    - The system would be able to alert drivers of emergency vehicles. This may take some amount of training a model though, so that is why it is under evaluation.
- **Out of Scope**
  - SYS-10: Complex Hazard Detection

        - Our system will not be able to predict vehicle accidents or determine hazardous vehicles (i.e. drunk drivers).
    - SYS-11: Traffic Detection
        - Our system is primarily focused on hazards. There are other, more efficient ways of determining whether there is traffic.
    - SYS-12: Driver Attention Detection
        - Our system will not use the front-facing smartphone camera to detect whether the driver is drowsy or not. This could hurt latency and increase cost.
    - SYS-13: Proprietary Mapping Software
        - Our system will not develop a new mapping software, but will instead rely upon existing APIs (i.e. Google Maps).
- **Assumption**
    - SYS-14: Legal Driver
        - Our system assumes that the driver will operate their vehicle legally, meaning that the driver has the proper equipment to mount the phone to their vehicle in a way that complies with their state's laws, that they have been trained and are licensed to operate a motor vehicle in their area, that they are of age to operate a motor vehicle, etc.
    - SYS-15: Phone User
        - Our system assumes that the driver will know and understand how to operate a smartphone, and that they understand the laws behind operating a phone in their jurisdiction. It also assumes they have interacted with GPS software before.
    - SYS-16: Internet Connectivity
        - Our system assumes that the driver will have internet connectivity while using the app.
- **Constraints**
    - SYS-17: AI Model Effectivity
        - Our system is constrained by pretrained computer vision models, so we may not be able to detect every single road hazard that exists (or reliably predict complicated road hazards, like accident prediction)
    - SYS-18: Monetary Constraint
        - Our system is constrained by the amount of money that we can spend on VLMs and cloud computing. Ultimately, we can get more accurate results with more money, but we have to balance our budget.

Software Requirements

- **FR-01: Pothole Detection**
  - **System Requirement IDs**: SYS-01, SYS-06
  - **Statement**: When the smartphone camera is active and the weather conditions are sufficiently clear, the system will analyze camera frames to identify visible potholes in the roadway.
  - **Rationale**: Automatic pothole detection is foundational to the RoadBuddy project, as it would allow the application to identify hazards without needing drivers to report them manually.
  - **Test Method**: Test the application by using a labeled dataset containing both images with and without potholes, captured under clear-weather conditions and testing for precision.
  - **Supporting Context**: A pothole is defined as a depression or hollow in a road surface caused by wear or sinking. Clear-weather conditions are defined as sunny days with no fog, precipitation, or other potential obscuring factors.
  - **Priority**: Must Have
- **FR-02: Hazard Classification**
  - **System Requirement IDs**: SYS-03, SYS-05, SYS-08
  - **Statement**: When a hazard is detected, the system will assign the detected hazard a severity level based on some predefined criteria based on the potential impact of the hazard on drivers and vehicles. Either a warning will be given or a reroute depending on the severity.
  - **Rationale**: Not every hazard requires drivers to take a different route. A severity classification lets RoadBuddy distinguish between different hazards in a meaningful manner.
  - **Test Method**: Test by using a labeled dataset containing potholes of different levels of severity and testing for classification accuracy.
  - **Supporting Context**: A hazard refers to anything on the road which could pose a risk for driver and vehicle safety.
  - **Priority**: Must Have
- **FR-03: Hazard Reporting**
  - **System Requirement IDs**: SYS-04, SYS-05, SYS-16
  - **Statement**: When a hazard is detected and location data is available, the system will create a hazard report and send the report to a shared hazard database.
  - **Rationale**: Sharing hazard reports allows for hazard information detected by one driver to be known by other drivers in the area. Real-time reporting gives useful and actionable information to other drivers.

    - **Test Method**: Test by simulating hazard detections with GPS and internet connectivity available. Verify that the reports fulfill all requirements and that the coordinates are accurate to the test location. Also verify the reports reach the database in a timely manner.
    - **Supporting Context**: A hazard report is a record containing all required information for sharing hazard data (time of discovery, location data, severity, etc.)
    - **Priority**: Must Have
- **FR-04: Driver Notifications**
    - **System Requirement IDs**: SYS-03, SYS-04, SYS-05
    - **Statement**: When a vehicle approaches a reported hazard within a certain specified distance along its current route, the system will notify the driver of the hazard's location and severity.
    - **Rationale**: Drivers need sufficiently advanced notice of reported hazards to be able to respond safely. Automatic alerts ensure drivers won't have to manually check a map for this information.
    - **Test Method**: Test by simulating a vehicle approaching a reported hazard and seeing if a notification is triggered within the specified distance with all the appropriate information.
    - **Supporting Context**: The specified distance refers to the distance ahead a hazard is along the current path, not just a general radius.
    - **Priority**: Must Have
- **FR-05: Camera Activation & Video Capture**
    - **System Requirement IDs**: SYS-01, SYS-04, SYS-06, SYS-15
    - **Statement**: When a user activates hazard detection, the system will access their smartphone's rear-facing camera to capture roadway frames for analysis.
    - **Rationale**: Camera input is necessary for hazard detection and foundational to RoadBuddy.
    - **Test Method**: Test by using hazard detection on supported smartphones and verifying that the camera is activated and recording, and frames are being captured for use by the detection and classification models.
    - **Supporting Context**: Frames refer to individual images as taken from the full live recording.
    - **Priority**: Must Have
- **FR-06: GPS Location Acquisition**
    - **System Requirement IDs:** SYS-02, SYS-04, SYS-16
    - **Statement:** When the user activates RoadBuddy navigation or hazard detection, the system shall obtain the smartphone's current GPS location and associate the most recent available coordinates with each detected hazard.

    - **Rationale:** Accurate location information is required so that RoadBuddy can store where hazards were detected, determine whether a hazard lies along another user's route, and provide appropriate warnings or rerouting.
    - **Test Method:** Activate RoadBuddy on a supported smartphone with location permissions enabled. Verify that the system retrieves the device's coordinates and that a simulated hazard report contains GPS coordinates within the accuracy range reported by the device.
    - **Supporting Context:** GPS location refers to the latitude and longitude supplied by the smartphone's location services. Each hazard report will use the most recent valid location available at the time of detection.
    - **Priority:** Must Have
- **FR-07: Google Account Authentication**
    - **System Requirement IDs:** SYS-15, SYS-16
    - **Statement:** When an unauthenticated user launches RoadBuddy, the system shall allow the user to authenticate using a Google account before accessing the primary application functions.
    - **Rationale:** Google authentication provides a familiar and relatively simple onboarding process while allowing RoadBuddy to associate application activity with an identified user account.
    - **Test Method:** Install the application on a test device with no existing RoadBuddy session. Sign in using a valid Google account and verify that authentication succeeds, the user is given access to the application, and the same RoadBuddy account is retrieved during a subsequent login.
    - **Supporting Context:** Authentication refers to verifying the identity of a user through Google's authentication service. RoadBuddy will rely on Google for authentication rather than creating and storing user passwords itself.
    - **Priority:** Must Have
- **FR-08: Local First-Stage Hazard Screening**
    - **System Requirement IDs:** SYS-01, SYS-04, SYS-17, SYS-18
    - **Statement:** While hazard detection is active, the system shall process selected roadway camera frames using a locally running computer vision model before requesting analysis from a cloud-based model.
    - **Rationale:** Performing the first stage of detection locally reduces latency, network usage, and cloud processing costs. It also allows RoadBuddy to eliminate frames that are unlikely to contain hazards before using a more expensive model.
    - **Test Method:** Run RoadBuddy using a labeled set of roadway video containing frames both with and without hazards. Verify that each sampled frame is first processed by the local model and that frames not identified as potential hazards are not automatically submitted to the cloud-based model.
    - **Supporting Context:** The local model is a computer vision model that executes directly on the smartphone. A candidate hazard is a frame that

meets the local model's configured detection threshold and therefore qualifies for additional processing.
    - **Priority:** Must Have
- **FR-09: Cloud VLM Hazard Verification**
    - **System Requirement IDs:** SYS-03, SYS-05, SYS-16, SYS-17, SYS-18
    - **Statement:** When the local computer vision model identifies a candidate hazard that meets the configured verification threshold, the system shall submit the candidate hazard to the cloud-based VLM for secondary classification and severity analysis.
    - **Rationale:** Selective use of a VLM allows RoadBuddy to use a more capable model for difficult classification decisions without requiring every camera frame to be processed in the cloud.
    - **Test Method:** Provide the local model with a labeled test set containing both candidate hazards and non-hazard frames. Verify that qualifying candidate hazards are submitted to the VLM, rejected frames are not submitted, and the VLM response is associated with the correct hazard record.
    - **Supporting Context:** VLM refers to a vision-language model capable of analyzing visual information and returning a classification. Sparse VLM use means that the cloud model is invoked only after the local model determines that additional analysis is warranted.
    - **Priority:** Must Have
- **FR-10: Mapping and Real-Time Rerouting**
    - **System Requirement IDs:** SYS-02, SYS-03, SYS-04, SYS-16
    - **Statement:** When a reported hazard with a severity requiring avoidance is located along the user's active route, the system shall request an updated route from the integrated mapping service that avoids the hazard when an alternate route is available.
    - **Rationale:** Severe hazards may require more than an audible warning. Integration with an existing mapping service allows RoadBuddy to provide updated navigation without developing proprietary mapping software.
    - **Test Method:** Create a simulated severe hazard on an active test route and verify that RoadBuddy requests a new route from the mapping service. Confirm that, when an alternate route is available, the updated route does not pass through the defined hazard location.
    - **Supporting Context:** A severe hazard is a hazard whose classification meets the predefined threshold for rerouting. The mapping service refers to a third-party mapping API, such as Google Maps, used for navigation and route generation.
    - **Priority:** Must Have
- **FR-11: Hazard Data Retrieval**
    - **System Requirement IDs:** SYS-02, SYS-04, SYS-16

    - **Statement:** When the user starts navigating, the system shall retrieve the reported hazards along the route taken by the user from the hazard database.
    - **Rationale:** RoadBuddy must have access to the database of previous reported hazards to inform the drivers of the road ahead.
    - **Test Method:** Create multiple simulated hazard reports in the database and start a navigation route to ensure that the system retrieves all the relevant hazards.
    - **Supporting Context:** The reported hazards refer to the database of the hazards detected and reported earlier and stored in a shared database.
    - **Priority:** Must have
- **FR-12: Duplicate Hazard Prevention**
    - **System Requirement IDs:** SYS-04, SYS-05
    - **Statement:** When a driver detects a hazard and reports it, the system shall check if the same hazard or a hazard at a similar location has been reported to avoid similar reports.
    - **Rationale:** Multiple drivers will likely detect the same hazard. Preventing duplicate reports will help keep the database organized and avoid unnecessary notifications.
    - **Test Method:** Create duplicate hazard reports by simulating multiple reports of the same hazard and ensure that there aren't duplicate entries.
    - **Supporting Context:** Duplicate Hazards refer to the hazard detection reports that are similar to the previous report.
    - **Priority:** Must have
- **FR-13: Hazard Report Expiration**
    - **System Requirement IDs:** SYS-04, SYS-05
    - **Statement:** When a hazard report exceeds a predefined expiration period, it shall be removed from the database unless a new report extends the duration.
    - **Rationale:** Road conditions might change, and the hazards earlier reported should be cleared. This will ensure that the system does not use stale information and keeps hazard reports fresh.
    - **Test Method:** Save multiple hazard reports with different expiry periods and ensure that the expired reports are removed while the active reports are displayed.
    - **Supporting Context:** The expiration period refers to the amount of time a hazard report remains active before requiring an update.
    - **Priority:** Must have
- **FR-14: Hazard Detection Controls**
    - **System Requirement IDs:** SYS-01, SYS-04, SYS-15

    - **Statement:** When the system is set to disable hazard detection, the system shall stop saving the camera feeds to the detection models and act as a read only mode.
    - **Rationale:** Hazard detection can be stopped by the user when they do not want it to run. This will allow the user to be able to protect their data if they do not want to upload their feed.
    - **Test Method:** Start the hazard detection system and ensure that the camera feeds are saved to the system and stop when the hazard detection is disabled.
    - **Supporting Context:** Hazard detection refers to the process of capturing video frames for processing using CV models in order to update the database.
    - **Priority:** Must have
- **FR-15: Hazard Detection Controls**
    - **System Requirement IDs:** SYS-02, SYS-03, SYS-05
    - **Statement:** When the route overview is opened, the system shall display and mark all the detected hazards along the route with their respective severity and route impact.
    - **Rationale:** Markers will inform the drivers of the location of hazards and help them know what to expect on the road ahead.
    - **Test Method:** Save multiple hazards with varying severity levels and ensure that the severity markers are displayed on the map where the hazard was detected.
    - **Supporting Context:** A map marker is a visual indicator used to represent the location of a reported hazard.
    - **Priority:** Must have

Non-Functional Requirements

- **NFR-01: Local Model Latency**
    - **System Requirement IDs**: SYS-04, SYS-17
    - **Statement**: While hazard detection is active on a supported smartphone, the local first-stage computer vision model shall return a detection result for at least 95% of sampled frames within 500 milliseconds of receiving the frame.
    - **Rationale**: RoadBuddy depends on rapid detection because a road hazard becomes less useful to drivers if significant time passes between observing the hazard and reporting it.
    - **Test Method**: Run at least 1,000 sampled roadway frames through the local model on the supported test smartphone. Record inference time for each frame and verify that at least 95% complete within 500 milliseconds.

    - **Supporting Context**: Model latency is measured from the time a sampled camera frame is provided to the local model until the model returns its detection result. Network and cloud processing times are not included in this measurement.
    - **Priority**: Must Have
- **NFR-02: Cloud Processing Efficiency**
    - **System Requirement IDs**: SYS-17, SYS-18
    - **Statement**: During normal clear-weather operation, the system shall process at least 90% of sampled roadway frames without requiring a cloud VLM request.
    - **Rationale**: Cloud model requests introduce both monetary cost and additional latency. RoadBuddy should therefore rely primarily on the local model and reserve VLM processing for frames that require additional analysis.
    - **Test Method**: Run RoadBuddy against a representative clear-weather roadway test set containing at least 1,000 sampled frames. Record the number of frames analyzed locally and the number submitted to the cloud VLM. Verify that no more than 10% of the sampled frames result in VLM requests.
    - **Supporting Context**: A cloud VLM request occurs whenever RoadBuddy transmits a candidate image to the cloud model for additional analysis. The 90% target measures use of the local processing stage rather than hazard-detection accuracy.
    - **Priority**: Must Have
- **NFR-03: User Onboarding Usability**
    - **System Requirement** IDs: SYS-15, SYS-16
    - **Statement**: When a new user has a valid Google account and grants the required application permissions, the user shall be able to complete authentication and reach the RoadBuddy navigation interface within 60 seconds.
    - **Rationale**: RoadBuddy is intended for use while preparing to drive, so account setup should require minimal time and interaction. A complicated onboarding process could discourage users from adopting the application.
    - **Test Method**: Conduct an onboarding test with at least 10 users who have not previously used RoadBuddy. Measure the time from selecting Google authentication until reaching the navigation interface and verify that each user can complete the process within 60 seconds without assistance.
    - **Supporting Context**: Onboarding includes Google authentication and required application permissions. Time spent creating a new Google account is not included.
    - **Priority**: Must Have

- **NFR-04: Detection Accuracy**
  - **System Requirement IDs:** SYS-01, SYS-05, SYS-06
  - **Statement:** When operating in clear weather conditions, the system shall have at least 85% precision and 80% recall in detecting potholes.
  - **Rationale:** Too many false positives may cause drivers to lose trust in the system while false negatives will create unsafe driving conditions. An accurate detection model is required for an improved user experience.
  - **Test Method:** On a test set of images containing potholes and non-potholes, calculate the precision and recall of the model and ensure that they are each above 85% and 80%.
  - **Supporting Context:** Precision measures how many detected potholes are actual potholes, while recall measures how many actual potholes the model successfully detects.
  - **Priority:** Must have
- **NFR-05: Hazard Reporting Reliability**
  - **System Requirement IDs:** SYS-04, SYS-05, SYS-16
  - **Statement:** When connected to the Internet, the system shall report at least 99% of the generated hazard reports to the shared database when operating in stable network conditions.
  - **Rationale:** The entire purpose of RoadBuddy is to warn other drivers of hazards. If the system fails to report hazards, then it is failing at its primary function.
  - **Test Method:** Generate 1,000 hazard reports in stable Internet connection conditions and ensure that at least 990 of them are reported to the shared database.
  - **Supporting Context:** A hazard report is successfully uploaded to the shared database if the database contains an entry of the uploaded report with identical contents.
  - **Priority:** Must Have

Deliverables

- **D-01: RoadBuddy Mobile Application**
  - A functional mobile Android application which uses a smartphone's rear-facing camera to capture footage of the road for detecting potholes and providing drivers with reliable hazard information.
  - **Requirements:** FR-01, FR-03, FR-04, FR-05, FR-06, FR-10, FR-14
- **D-02: Hazard Notification Interface**
  - An interface which presents relevant hazard warning information to drivers approaching reported potholes or other hazards.

    - **Requirements:** FR-02, FR-03, FR-04, FR-06, FR-10, FR-14
- **D-03: Hazard Reporting Database**
    - A functioning database to store hazard reports for later use.
    - **Requirements:** FR-03, FR-04, FR-10, FR-11, FR-12, FR-13, NFR-05
- **D-04: Computer Vision/VLM Instance**
    - A trained model for detecting potholes and other hazards in real-time.
    - **Requirements:** FR-01, FR-02, FR-05, FR-08, FR-09, NFR-01, NFR-02, NFR-04
- **D-05: Software Verification/Testing Report**
    - A report documenting test procedures, datasets, expected results, actual results, and problem areas for RoadBuddy. Will provide a quantitative look at RoadBuddy's effectiveness.
    - **Requirements:** FR-01, FR-02, FR-03, FR-05, FR-06, FR-08, FR-09, FR-10, FR-11, FR-12, FR-13, NFR-01, NFR-02, NFR-04
- **D-06: End-User & Developer Documentation**
    - Documentation package for both end-users and developers. End-user documentation includes an installation and starting guide. Developer documentation includes code, architecture diagrams, etc.
    - **Requirements:** FR-05, FR-07, NFR-03

Standards

1. **ISO/IEC 25010:2023 – Software Product Quality**

URL: https://webstore.iec.ch/en/publication/90024

Applicability: RoadBuddy must detect hazards reasonably quickly, but does not have to fully real-time. This standard applies because it establishes key software quality characteristics including performance, reliability, and usability.
Impact: We will focus on efficiently processing camera frames, reliably identifying/classifying hazards, and providing user interface that minimizes user input to reduce distractions on roadways.
Compliance: We will test our ability to correctly recognize hazards in a specified amount of time to ensure adherence to our software quality criteria.

2. **ISO/IEC 42001:2023 – Artificial Intelligence Management Systems**

URL: https://csrc.nist.gov/pubs/sp/800/218/final

Applicability: Because RoadBuddy tracks GPS coordinates and communicates with cloud services, there is potential for security vulnerabilities within the software. The SSDF provides a framework for mitigating these risks by developing the software in a secure manner.
Impact: We will incorporate security into our design process by thinking about secure API authentication when communicating with web services. We will also make sure to use a secure database when storing any sensitive user information.
Compliance: We will test compliance by thoroughly reviewing code for any security vulnerabilities and will do penetration testing to ensure none of our services can be accessed by malicious actors.

3. **ISO/IEC 42001:2023 – Artificial Intelligence Management Systems**

URL: https://www.iso.org/standard/42001

Applicability: RoadBuddy utilizes both local computer vision models and remote VLM models to efficiently detect hazards. This standard is relevant because we have to manage the nondeterministic nature of CV models and make sure to avoid negatively affecting routing from false detections.
Impact: In designing the AI hazard detection system, we must acknowledge any limitations associated with each model and prepare our implementation for scenarios where the prediction is inaccurate.
Compliance: We will evaluate our models using real-world data to assess limitations. We will use this evidence to design a robust system and minimize error.

Gen AI statement: Generative AI was used for general brainstorming ideation. For functional and non-functional requirements generative AI was used to make sure all bullet points were sufficient for each requirement.